$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}\,\dagger & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}\,\dagger^{\alpha\beta\chi} & 0 & Y_1 k^2 + 3\overset{(2)}{\kappa}_1
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}\,\dagger^{\alpha\beta} & \frac{1}{2}\left(Y_2 k^2 + 4\overset{(2)}{\kappa}_1\right) & \frac{Y_2 k^2+4\overset{(2)}{\kappa}_1}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}\,\dagger^{\alpha\beta} & \frac{Y_2 k^2+4\overset{(2)}{\kappa}_1}{2\sqrt{2}} & \frac{1}{4}\left(Y_2 k^2 + 4\overset{(2)}{\kappa}_1\right) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}\,\dagger^{\alpha} & 0 & 0 & \frac{Y_3 k^2}{6} & -\frac{Y_3 k^2}{6\sqrt{2}} \\
\mathcal{K}^{\#2}_{1^-}\,\dagger^{\alpha} & 0 & 0 & -\frac{Y_3 k^2}{6\sqrt{2}} & \frac{Y_3 k^2}{12}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}\,\dagger^{\alpha\beta} & \frac{Y_4 k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}\,\dagger^{\alpha\beta\chi} & 0 & \frac{Y_4 k^2}{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}\,\dagger & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}\,\dagger^{\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(0^-)}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}\,\dagger^{\alpha\beta} & \frac{2}{3\,\mathrm{Det}(1^+)} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}\,\dagger^{\alpha\beta} & \frac{\sqrt{2}}{3\,\mathrm{Det}(1^+)} & \frac{1}{3\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}\,\dagger^{\alpha} & 0 & 0 & \frac{2}{3\,\mathrm{Det}(1^-)} & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}\,\dagger^{\alpha} & 0 & 0 & -\frac{\sqrt{2}}{3\,\mathrm{Det}(1^-)} & \frac{1}{3\,\mathrm{Det}(1^-)}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}\,\dagger^{\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}\,\dagger^{\alpha\beta\chi} & 0 & \frac{1}{\mathrm{Det}(2^-)}
\end{array}
$$

Abbreviations used in matrices

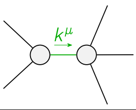
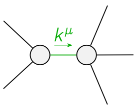
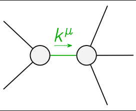
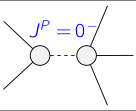
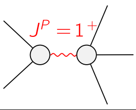
$$
Y_1 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \;\&\&\; Y_2 == 2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - 2\,\overset{(4)}{\kappa}_{3} \;\&\&
$$
$$
Y_3 == 6\,\overset{(4)}{\kappa}_{1} + 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \;\&\&\; Y_4 == 2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \;\&\&\; \mathrm{Det}(0^-) == k^2\left(\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16}\right) + 3\,\overset{(2)}{\kappa}_{1} \;\&\&
$$
$$
\mathrm{Det}(1^+) == \frac{3}{4}\,k^2\left(2\,\overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - 2\,\overset{(4)}{\kappa}_{3}\right) + 3\,\overset{(2)}{\kappa}_{1} \;\&\&\; \mathrm{Det}(2^+) == \frac{1}{2}\,k^2\left(2\,\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}\right)
$$

Lagrangian

$$
\overset{(2)}{\kappa}_1\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(2)}{\kappa}_1\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} -
$$
$$
\overset{(4)}{\kappa}_1\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \frac{2}{3}\,\overset{(4)}{\kappa}_{15}\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \frac{1}{3}\,\overset{(4)}{\kappa}_{16}\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} +
$$
$$
\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + 2\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} + \overset{(4)}{\kappa}_{16}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - 2\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} -
$$
$$
\overset{(4)}{\kappa}_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 2\,\overset{(4)}{\kappa}_{15}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \overset{(4)}{\kappa}_{16}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{1}{2}\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} -
$$
$$
\frac{1}{4}\,\overset{(4)}{\kappa}_{16}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} - \frac{1}{2}\,\overset{(4)}{\kappa}_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} + 2\,\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\,\alpha\beta}_{1^+} == \mathcal{J}^{\#2\,\alpha\beta}_{1^+}$	3	$3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} + 2\,\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$
Total # constraint(s):	7	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{1}{6\,\overset{(4)}{\kappa}_1+2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}}$
	2	0	$-\frac{1}{2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}}$
	2	0	$\frac{1}{2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}}$
	1	$\frac{3\,\overset{(2)}{\kappa}_1}{-\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}}$	$\frac{1}{-\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}}$
	3	$\frac{4\,\overset{(2)}{\kappa}_1}{-2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{16}+2\,\overset{(4)}{\kappa}_3}$	$\frac{4}{6\,\overset{(4)}{\kappa}_{15}-3\left(\overset{(4)}{\kappa}_{16}+2\,\overset{(4)}{\kappa}_3\right)}$

Resolved unitarity condition(s):	(Demonstrably impossible)